# Mark Farinas

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## Education

**MacEwan University** *Jan 2024 – Present*
*BSc in Computer Science*

- **Relevant Coursework:** Data Structures and Algorithms, Coding in Python, Programming using C and UNIX, Linear Algebra, Human-Computer Interaction, Software Engineering
- **Award (Apr 2026):** 1st Place - CMPT 395 Software Engineering Competition (IEEE-Sponsored) - won both the Semi-Finals and the Grand-Finals with Schedulater
- **Publication (May 2026):** Co-authored full research paper submitted to IEEE SmartEdu 2026 - "Unjamming Deferrals: Supporting Students Through JAMN Bot, a Mobile RAG-Grounded Policy Assistant"

**University of Alberta** *Sept 2016 – April 2021*
*BSc in Immunology and Infection*

## Experience

**Alberta Federation of Rural Water Coops** *May 2026 – Present*
*Database Administrator Intern*

- Migrating a legacy Microsoft Access database for Alberta rural water co-operatives to a fully functional web-based database, modernizing data access for the Federation and its member co-ops
- Administrating the Water Federation website and automating standard workplace procedures, including form development and recurring administrative workflows
- Working across the stack with Python, SQL, HTML/CSS, and JavaScript; coordinating with stakeholders across rural Alberta water systems on data structure, access requirements, and rollout

## Skill Highlights

Android & Mobile Development

- Contributed to frontend screens for Schedulater in Kotlin + Jetpack Compose, including role-specific interfaces for students, faculty, and admin across a deployed multi-role platform
- Implemented the Android frontend for VocabMaxxing (login, dashboard, daily practice with microphone capture) in Kotlin + Jetpack Compose
- Integrated Retrofit REST client, Firebase Authentication, FCM push notification handling, and DataStore session management in Schedulater's Android client

Backend Development

- Architected Ktor + JetBrains Exposed + SQLite backend for Schedulater with JWT + jBCrypt auth, role-based route protection for three user types, and a room-booking engine with conflict detection and proctor load-balancing
- Built the Ktor + JetBrains Exposed + SQLite backend for VocabMaxxing with Firebase Admin token verification, vocabulary word seeding, and REST endpoints for auth, attempts, and user dashboards
- Designed automated FCM push notification triggers via Firebase Admin SDK and a scheduled no-show poller in Schedulater's backend service layer
- Containerized both backends with Docker and deployed on Railway.io; configured environment-driven secrets management for AI, email, and auth services

AI Integration

- Built an Ollama-powered RAG pipeline for Schedulater's AI policy assistant - custom document chunker, nomic-embed-text vector embeddings, cosine-similarity retrieval over a SQLite vector store, and a role-aware prompt builder over ingested MacEwan policy PDFs (Apache PDFBox)

- Designed an OpenAI GPT-4o-mini semantic scoring service for VocabMaxxing with a multi-dimensional rubric (contextual usage, grammar, sentence complexity) and hardened system prompt defenses against prompt-injection, language, and content-policy attacks
- Developed a custom algorithmic scoring engine combining morphological stemming, structural-complexity heuristics, vocabulary diversity ratio, and grammar checks; validated with unit tests and combined with the AI scorer for a hybrid evaluation pipeline

UI/UX Design

- Led ideation and UX design for VocabMaxxing; produced low- and mid-fidelity Figma wireframes across multiple iterations covering layout, colour scheme, navigation flow, and gamification mechanics (streaks, XP, coin shop)
- Ideated and co-designed TiMED (Hack4Health 2025), a hardware-software medication management system; designed companion app wireframes featuring daily medication schedules, adherence reports, and QR-based device pairing; presented proof-of-concept to a panel of judges
- Developed a personal portfolio website using Next.js with a design system inspired by American Psycho's business card aesthetic, featuring client-side routing, responsive navigation, and interactive hover states

Research & Data Analysis

- Co-authored a research paper submitted to IEEE SmartEdu 2026 evaluating a RAG-grounded LLM policy assistant; designed a multi-condition benchmark dataset, ran temperature ablation studies, and computed inter-rater agreement between human evaluators and an LLM-Judge
- Built Calorie Tracker, a Python CLI application using SciPy non-linear optimization to estimate maintenance calories, generate personalized nutrition plans with algebraic and curve-fitted goal recommendations, and track weight trends via moving averages
- Conducted COVID-19 surveillance research at Alberta Precision Labs (Dec 2022 – Dec 2023) using qPCR and dPCR detection methods across multiple Albertan communities; executed daily lab workflows including RNA extraction, RT-PCR, and digital PCR
- Developed Audio-to-Display, a Python audio processing tool using Whisper AI to transcribe and parse automated phone menu prompts into clean, formatted output

Communication & Teamwork

- Served as Backend Developer on a 5-person Agile/Scrum team building Schedulater; participated in bi-weekly sprints, code reviews, and cross-team conflict resolution - project won 1st Place at the IEEE-sponsored CMPT 395 Grand-Finals
- Provide solution-based customer service at Popeye's Supplements (July 2022 – Present), advising clients on supplements for athletic performance and wellness goals; conduct in-store demos and community education initiatives
- Performed data analysis and maintained detailed lab records at Alberta Precision Labs, supporting research accuracy and inter-team communication across COVID-19 surveillance workflows